Henrique Assumpção

Belo Horizonte, Brazil
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Education

MS in Computer Science Mar 2025 – Jul 2026
Federal University of Minas Gerais, Brazil *GPA: 98%*
- **Thesis**: Semidefinite Optimization on Laplacians of Highly Regular Graphs
- **Advisor**: Prof. Gabriel Coutinho

BS in Computer Science Mar 2020 – Dec 2024
Federal University of Minas Gerais, Brazil *GPA: 94%*
- **Thesis**: Algebras, Groups, and Graphs
- **Advisor**: Prof. Gabriel Coutinho
- **Minor**: Mathematics

Publications

- **Henrique Assumpção**, Gabriel Coutinho, Chris Godsil. Total Conformal Rigidity in Graphs. *Submitted*, 2026.
- **Henrique Assumpção**, Diego Ferreira, Leandro Campos, Fabricio Murai. CodeEvolve: an open-source evolutionary coding agent for algorithmic optimization. *Submitted*, 2026.
- **Henrique Assumpção**, Gabriel Coutinho. Semidefinite programming bounds on fractional cut-cover and maximum 2-SAT for highly regular graphs. *Mathematics of Operations Research*, 2026.
- **Henrique S. Assumpção**, Fabrício Souza, Leandro Lacerda Campos, Vinícius T. de Castro Pires, Paulo M. Laurentys de Almeida, Fabricio Murai. DELATOR: Money Laundering Detection via Multi-Task Learning on Large Transaction Graphs. *IEEE International Conference on Big Data* (IEEE BigData 2022).
- Bárbara Silveira, **Henrique S. Silva**, Fabricio Murai, Ana Paula C. da Silva. Predicting user emotional tone in mental disorder online communities. *Future Generation Computer Systems*, 2021.

Professional Experience

Machine Learning Research Scientist Jun 2025 - Ongoing
Inter Science, Brasil *Full Time*
- Created CodeEvolve, an open-source evolutionary coding agent inspired by AlphaEvolve, capable of achieving state-of-the-art results across multiple algorithmic benchmarks.
- Lead developer on the *LTMs* project: large-scale training of LLMs on transactional data for credit analysis and recommendation, covering architecture design, pre-training, inference optimization, and distributed training pipelines.

Undergraduate Researcher Aug 2021 - Feb 2022
Inter/DCC-UFMG, Brasil *Internship*
- Developed *DELATOR*, a scalable Graph Neural Network system for money laundering detection on large transaction graphs. Our results were published at *IEEE Big Data 2022*.
- Deployed models on graphs with 20M+ nodes and 100M+ edges, uncovering suspicious activity missed by rule-based baselines.

Data Science Instructor Jun 2022 - Dec 2022
Usiminas/DCC-UFMG, Brasil *Part time*
- Designed and delivered a comprehensive curriculum on Python, Pandas, and PyTorch, upskilling engineering teams at a major steel manufacturer to adopt data-driven workflows.
- Mentored multidisciplinary teams in building and deploying internal ML tools, resulting in the automation of manual data analysis tasks.

A.I. Research & Development — Mar 2021 - Aug 2021
Plus Three, USA — *Part Time*

- Engineered end-to-end NLP pipelines for automated question-answering and text generation, integrating transformer-based models into client-facing web applications to enhance chatbot capabilities.
- Authored a series of technical articles on AI ethics and bias for the nonprofit *AIandYou*, translating complex algorithmic concepts for a broad non-technical audience.

Research Projects

Quantum Computing for Finance — Jan 2025 - Jun 2025
Inter/DCC-UFMG, Brasil

- Implemented Quantum Support Vector Machines (QSVM) using Qiskit to benchmark quantum kernel methods against classical SOTA baselines for high-dimensional fraud detection.
- Engineered data encoding circuits to map complex financial features into quantum states, evaluating potential quantum advantage in risk modeling and classification accuracy.

Optimization & Spectral Graph Theory — Mar 2023 – Jan 2025
DCC-UFMG, Brasil

- Derived semidefinite programming (SDP) bounds for NP-hard problems (MAXCUT, MAX 2-SAT), providing theoretical grounding for complex optimization tasks.
- Leveraged spectral methods to analyze structured representations in clinical brain networks.

Predictive Maintenance for Steel Machinery — May 2021 - Jul 2021
MINASLIGAS/DCC-UFMG, Brasil

- Developed a Variational Autoencoder (VAE) in PyTorch for anomaly detection in multi-modal time-series telemetry, improving failure prediction by 10%.
- Deployed a real-time monitoring dashboard, translating deep learning insights into actionable industrial automation.

Sentiment Analysis on Online Mental Health Communities — Dec 2020 - May 2021
DCC-UFMG, Brasil

- Designed and optimized a novel RNN-based architecture in PyTorch to detect longitudinal shifts in emotional tone within large-scale online communities.
- Achieved a 20% performance increase over established baselines, leading to a publication in the journal Future Generation Computer Systems.

Talks and Presentations

- CodeEvolve: automated algorithmic discovery with LLMs. Invited Talk. AIDDA 2026, Remote.
- Gauge duality for parameters of highly regular graphs. Invited Talk. ILAS 2026, Blacksburg, USA.
- Evolutionary Coding Agents. Invited Talk. Automated Discovery at Scale 2026, San Francisco, USA.
- MAXCUT, Association Schemes and Semidefinite Programs. Invited Talk. CNMAC 2025, Rio de Janeiro, Brasil.
- Primal-dual inequalities for the approximate maximum-cut value of Graphs in Homogeneous Coherent Configurations. Poster Presentation. CNMAC 2024, Porto de Galinhas, Brasil.
- DELATOR: Money Laundering Detection via Multi-Task Learning on Large Transaction Graphs. Paper Presentation. IEEE BigData 2022, Osaka, Japan.

Technical Skills

Languages: Python, C++, C, CUDA, Java, JavaScript, Rust
ML & NLP: PyTorch, JAX, Triton, TensorFlow, HuggingFace, Scikit-Learn
Systems: Docker, AWS, Spark, Linux
Languages: Portuguese (Native), English (C2), Spanish (B2), French (B1)